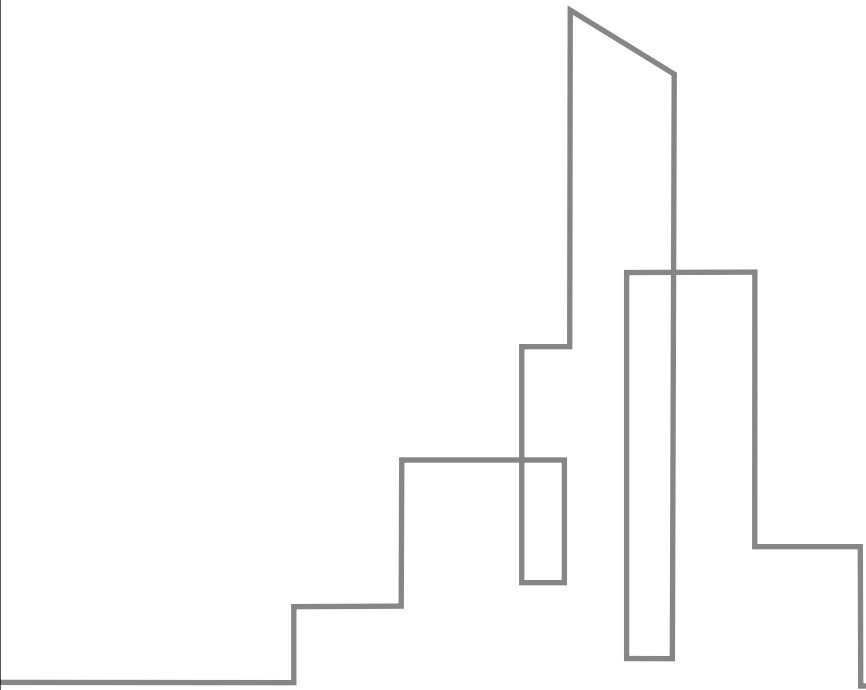


Prompt Engineering Guide



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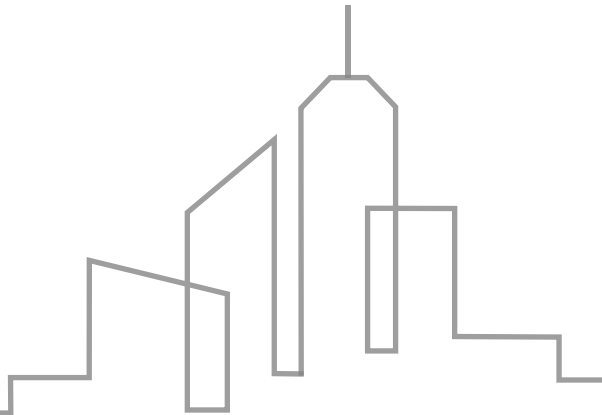
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SaaSCV Best
Practices



01

Introduction to Prompting



Prompting as core science



Prompt Engineering as core science

Prompt Engineering is the fundamental approach to achieving predictable and accurate outputs in saascv solutions.



Achieving predictable outputs

Prompt Engineering is essential for ensuring that saascv solutions produce consistent and reliable results.

Achieving predictable outputs

Output Length Management

Output length is managed to control cost and latency, not style, which is crucial for saascv solutions.

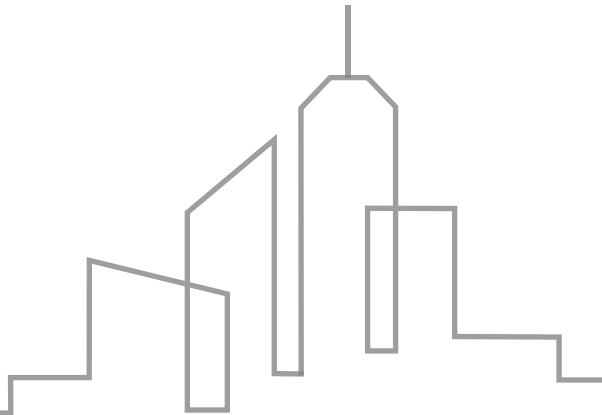
Sampling Controls

Sampling controls, including temperature and top K / top P, are used to ensure factual consistency and advanced randomness control in saascv solutions.



02

Controlling AI Engine



Output length management



Max Tokens for Cost Control

Used to manage cost and latency, not for altering the style of the output.



Efficient Resource Utilization

Ensures that the AI engine operates within resource constraints, optimizing performance.

Sampling controls for accuracy

01

Temperature Setting for Consistency

Low temperature settings are crucial for maintaining factual consistency in saascv.

02

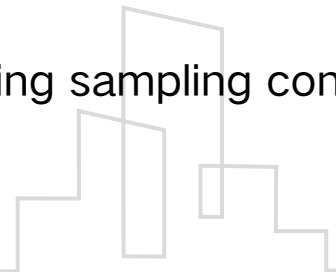
Top K and Top P Sampling

Advanced randomness controls that allow for fine-tuning of the AI's responses.

03

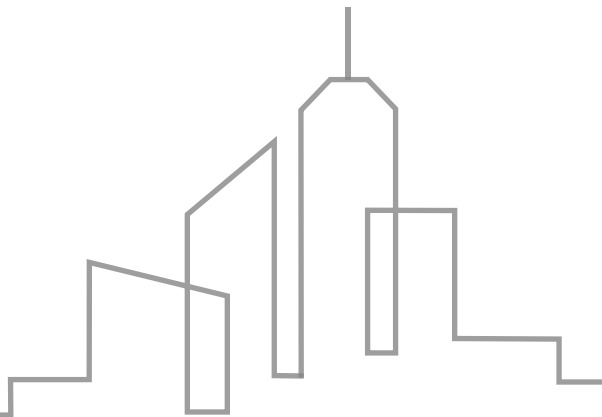
Balancing Cost and Accuracy

Trade-off between high accuracy and potential increased cost/latency when using sampling controls.



03

Core Prompting Techniques



Few-shot prompting for format

Providing examples in the prompt

Few-shot prompting involves giving examples in the prompt to guide the model into a specific format, such as JSON structure.



Forcing adherence to format

This technique ensures the AI adheres to the required format by showing it examples of the desired output.



System & role prompting



Setting the model's identity

System & role prompting involves defining the model's role or identity, like "Act as a Senior Developer," to elicit domain-specific responses.



Ensuring domain-specific responses

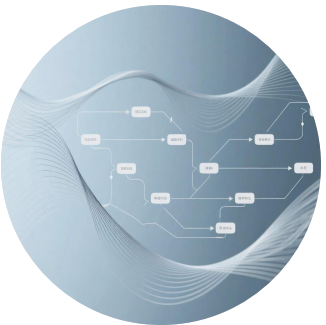
This method helps in tailoring the AI's responses to fit the context and requirements of a specific domain or role.

Chain-of-thought for logic



Improving accuracy for logic tasks

Chain-of-thought prompting adds a "Think step by step" directive to enhance the accuracy of responses in logic, math, and analysis tasks.



Facilitating step-by-step reasoning

This technique guides the AI to break down complex problems into a series of logical steps, improving the quality of its reasoning.



Advanced reasoning methods

01

Step-Back Prompting technique

Asking a general question first to enrich context before diving into specifics, which is part of advanced reasoning methods.

02

Self-Consistency approach

Using multiple runs and majority voting to achieve high accuracy, though it may increase cost and latency.

03

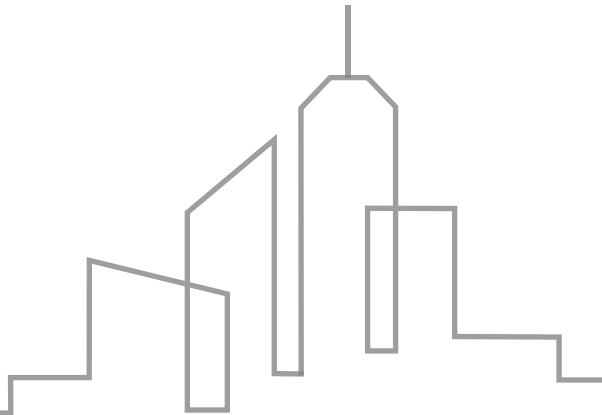
Trade-off considerations

Advanced reasoning methods require balancing accuracy with potential increases in cost and latency.



04

Agents and Real-World Interaction



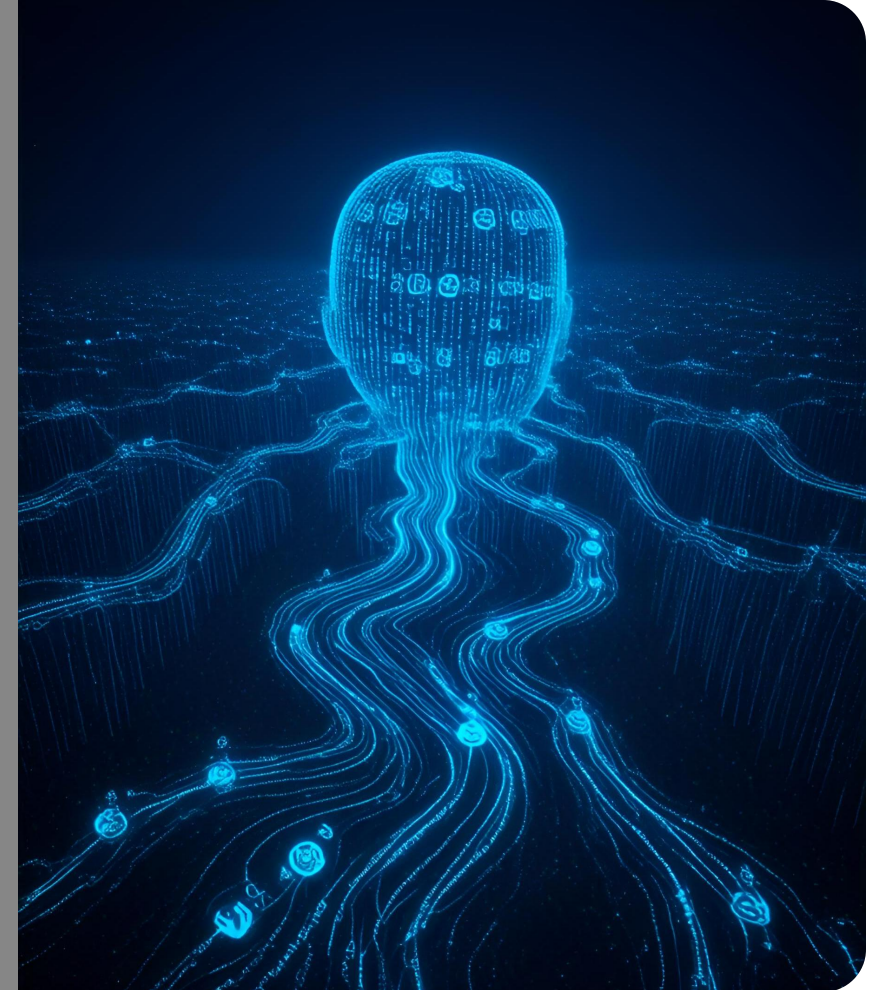
ReAct for intelligent agents

Combining reasoning with tools

ReAct integrates reasoning with external tools like search and code interpreter to enhance AI agents.

Creating intelligent AI agents

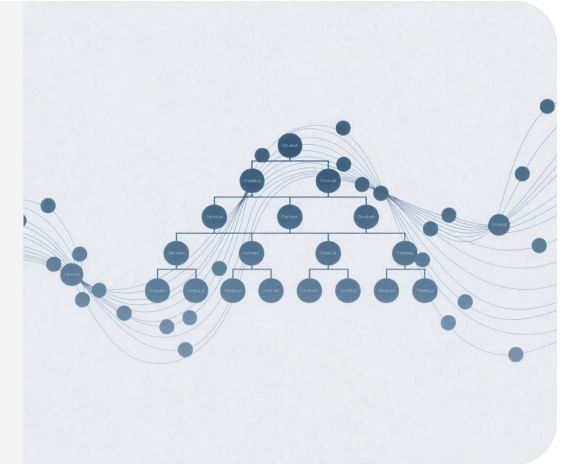
The method aims to develop AI agents that can reason and act effectively in real-world scenarios.



Tree-of-thought for complex problems

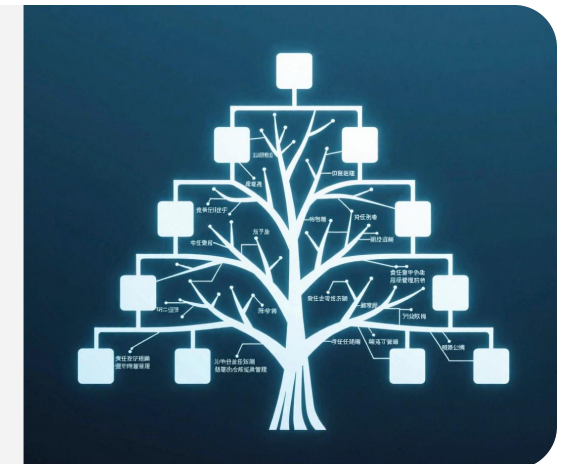
Exploring multiple reasoning paths

The Tree-of-Thought method examines various paths to solve complex problems, requiring a coding framework.



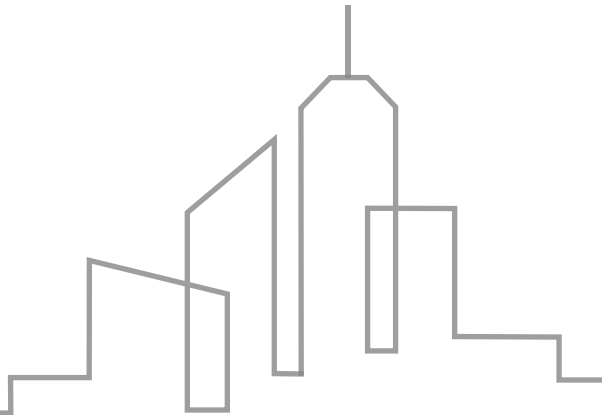
Complex problem-solving approach

ToT is specifically designed for tackling intricate issues by considering multiple logical sequences.



05

SaaSVC Best Practices



Clarity in design



Design with simplicity

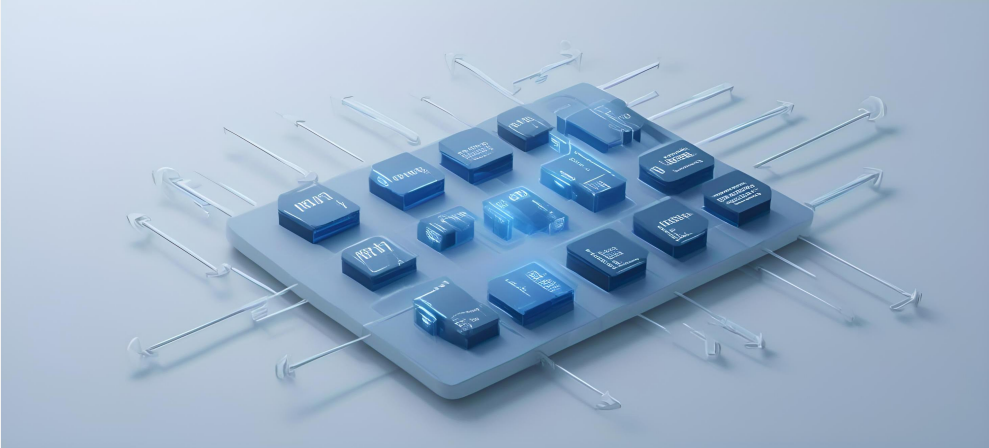
Clarity in design means creating interfaces and prompts with simplicity, making every requirement explicit.

Explicit requirements

Being explicit about every requirement ensures that the AI engine understands and executes tasks without ambiguity.

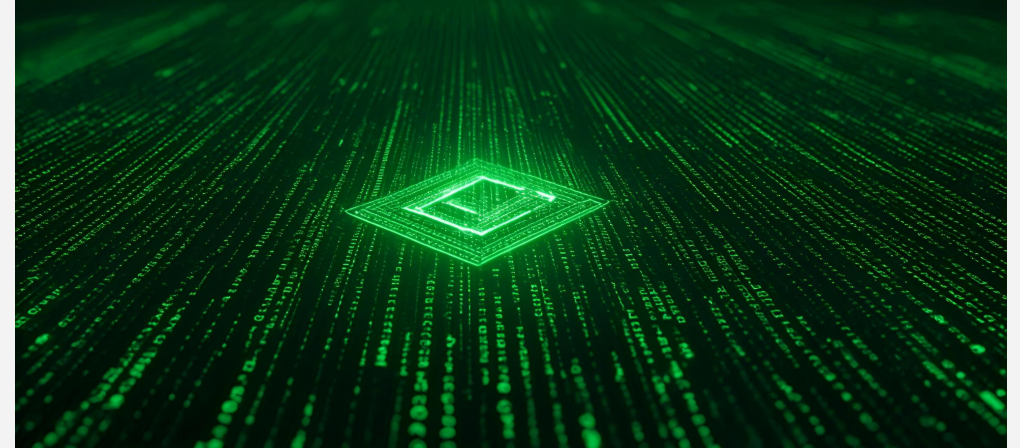


Specificity in instructions



Use instructions over constraints

Instructions should guide the AI, detailing the desired output format and behavior, rather than using constraints.



Specify output format

Clearly specify the output format, such as "valid JSON," to ensure the AI produces the expected results.

Accuracy with code prompting

Prompting with Code

Explicitly ask the model to write and execute code for calculation/data tasks to enhance accuracy.

Code execution for tasks

Using code prompting ensures the AI performs computational tasks with precision, reducing errors.

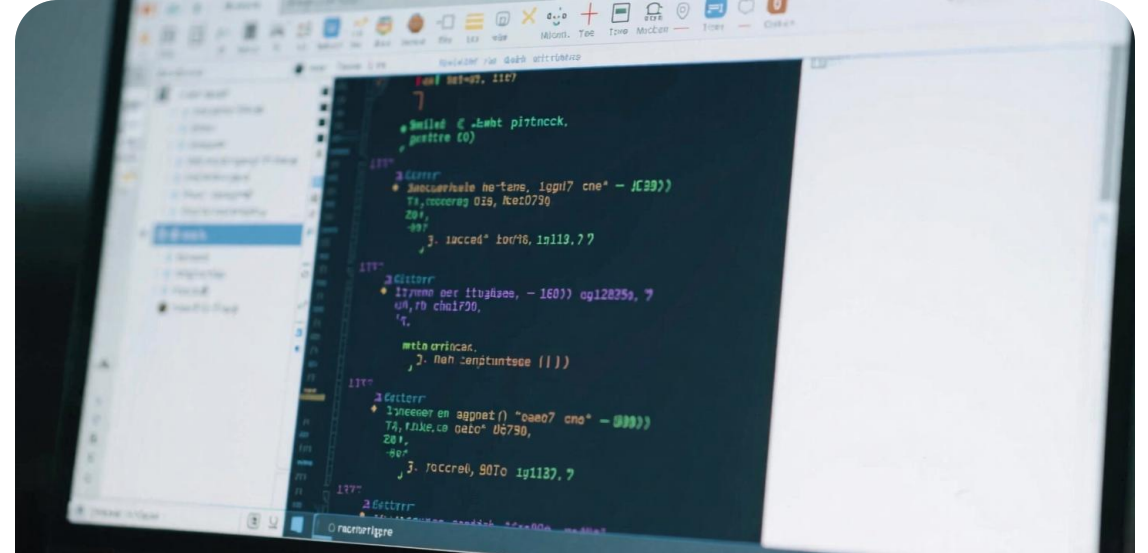


Deployment with variables



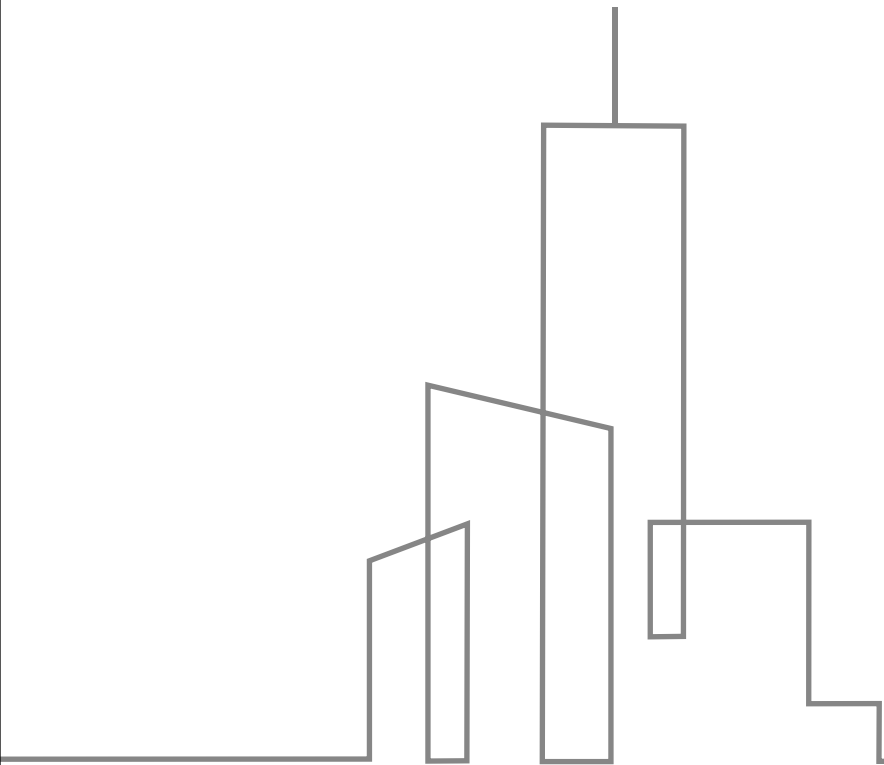
Use Variables in prompts

Incorporate variables in prompts for consistent, high-scale programmatic insertion of data.



Programmatic data insertion

This approach allows for the efficient and accurate deployment of AI solutions across various data sets.



THE END

Thank You